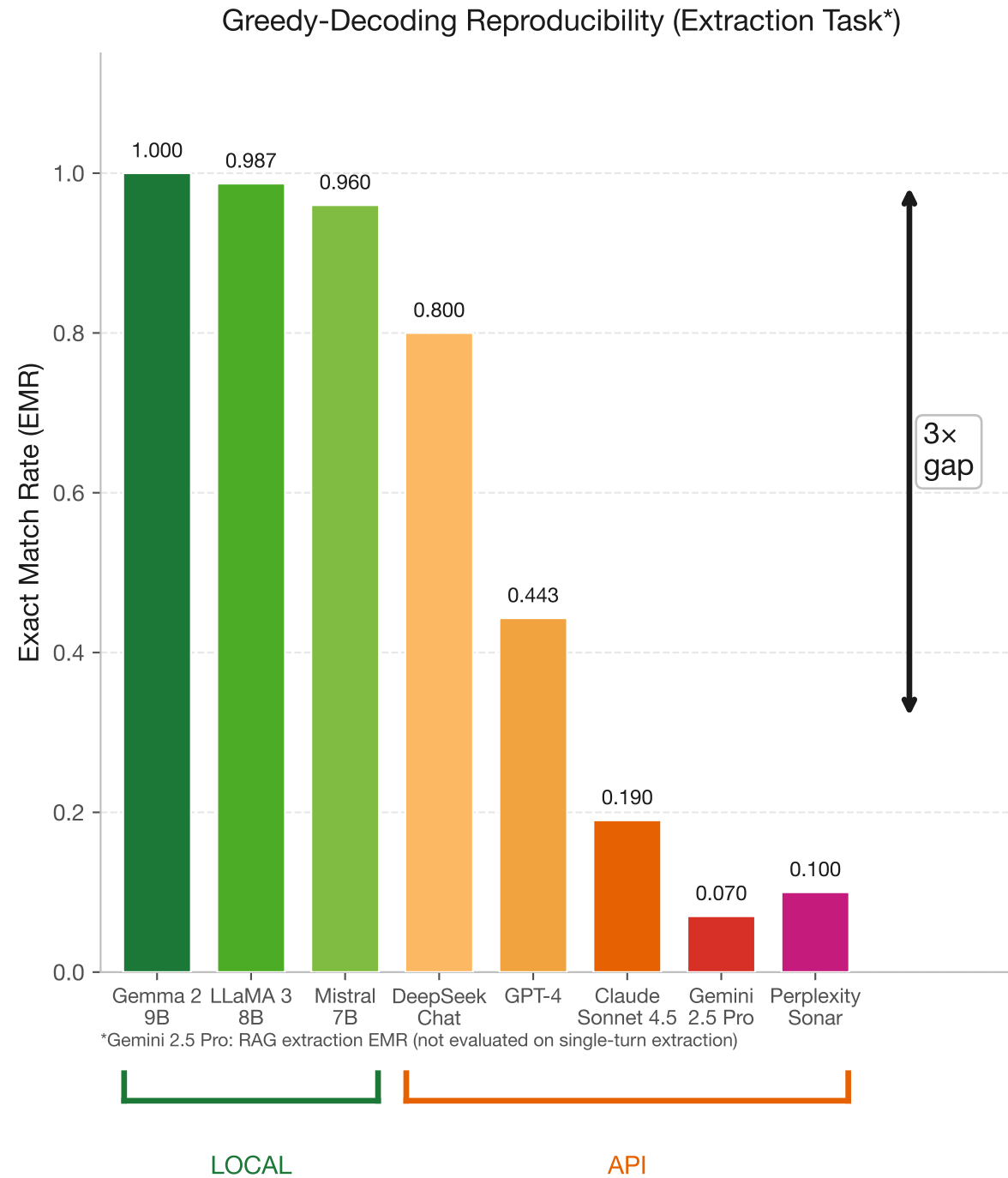
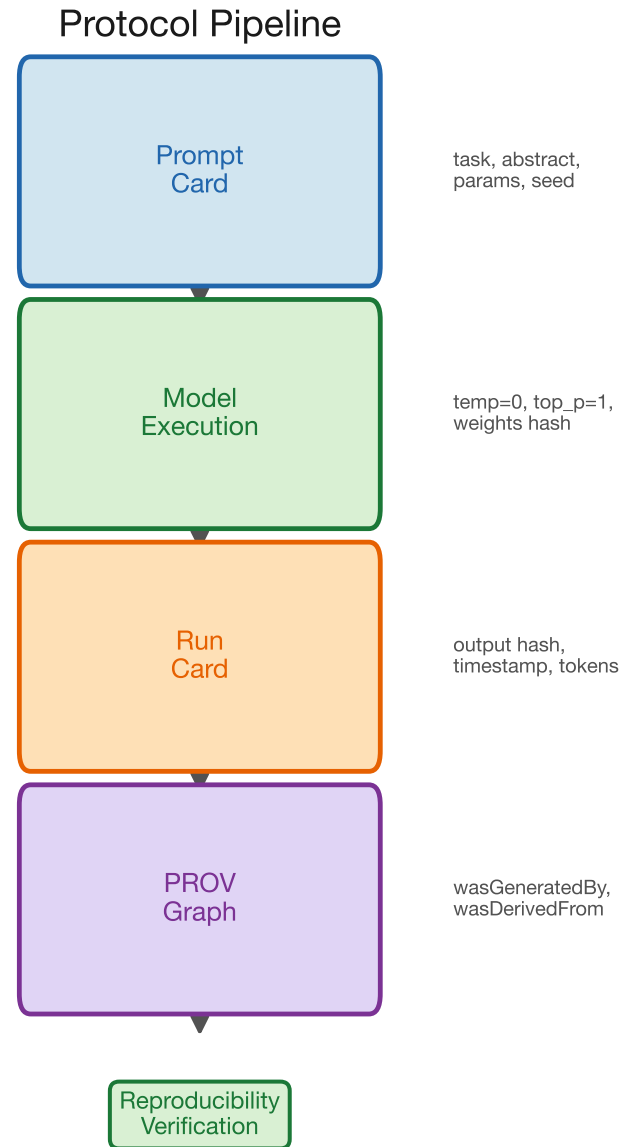


Hidden Non-Determinism in LLM APIs

A Lightweight Provenance Protocol for Reproducible Generative AI Research



*Gemini 2.5 Pro: RAG extraction EMR (not evaluated on single-turn extraction)

avg EMR = 0.982

avg EMR = 0.321

github.com/Roverlucas/genai-reproducibility-protocol

Key Results

7,004

total runs

9

deployment stacks

6

task families

3.1x

local-API gap

<1%

overhead

Local models reproduce near-perfectly; API outputs vary across identical calls.